

Adding and Subtracting Polynomials

Watch Khan Academy's **Adding polynomials** video at <https://youtu.be/ahdKdxsTj8E>

When adding two monomials of the same degree, you sum their coefficients:

$$7x^3 + 4x^3 = 11x^3$$

Using this idea, when adding two polynomials, you convert them into one long polynomial, then simplify by combining terms with the same degree. For example:

$$\begin{aligned}(10x^3 - 2x + 13) + (-5x^2 + 7x - 12) \\&= 10x^3 + (-2)x + 13 + (-5)x^2 + 7x + (-12) \\&= 10x^3 + (-5)x^2 + (-2 + 7)x + (13 - 12) \\&= 10x^3 - 5x^2 + 5x + 1\end{aligned}$$

Exercise 1 Adding Polynomials Practice

Add the following polynomials:

Working Space

1. $2x^3 - 5x^2 + 3x - 9$ and $x^3 - 2x^2 - 2x - 9$

2. $3x^5 - 5x^3 + 3x^2 - x - 3$ and $2x^4 - 2x^3 - 2x^2 + x - 9$

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Notice that in the second question, the degree 1 term disappears completely: $(-x) + x = 0$

One more tricky thing can happen — sometimes the coefficients don't add nicely. For example:

$$\pi x^2 - 3x^2 = (\pi - 3)x^2$$

That is as far as you can simplify it.

1.1 Subtraction

Now, watch Khan Academy's **Subtracting polynomials** at <https://youtu.be/5ZdxnFspyP8>.

When subtracting one polynomial from the other, it is a lot like adding two polynomials. The difference: when making the two polynomials into one long polynomial, we multiply each monomial that is being subtracted by -1. For example:

$$\begin{aligned}(2x^2 - 3x + 9) - (5x^2 - 7x + 4) \\&= 2x^2 + (-3)x + 9 + (-5)x^2 + 7x + (-4) \\&= (2 - 5)x^2 + (-3 + 7)x + (9 - 4) \\&= -3x^2 + 4x + 5\end{aligned}$$

Exercise 2 Subtracting Polynomials Practice

Subtract the following polynomials:

Working Space

$$1. (2x^3 - 5x^2 + 3x - 9) - (x^3 - 2x^2 - 2x - 9)$$

$$2. (3x^5 - 5x^3 + 3x^2 - x - 3) - (2x^4 - 2x^3 - 2x^2 + x - 9)$$

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1.2 Adding Polynomials in Python

As a reminder, in our Python code, we are representing a polynomial with a list of coefficients. The first coefficient is the constant term. The last coefficient is the leading coefficient. So, we can imagine $-5x^3 + 3x^2 - 4x + 9$ and $2x^3 + 4x^2 - 9$ would look like this: *FIXME: Diagram here*

To add the two polynomials then, we sum the coefficients for each degree. *FIXME: Diagram here*

Create a file called `add_polynomials.py`, and type in the following:

```

1  def add_polynomials(a, b):
2      degree_of_result = len(a)
3      result = []
4      for i in range(degree_of_result):
5          coefficient_a = a[i]
6          coefficient_b = b[i]
7          result.append(coefficient_a + coefficient_b)
8      return result
9
10 polynomial1 = [9.0, -4.0, 3.0, -5.0]
```

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```
11 polynomial2 = [-9.0, 0.0, 4.0, 2.0]
12 polynomial3 = add_polynomials(polynomial1, polynomial2)
13
14 print('Sum =', polynomial3)
```

Run the program.

Unfortunately, this code only works if the polynomials are the same length. For example, try making polynomial1 have a larger degree than polynomial2:

```
1 #  $x^4 - 5x^3 + 3x^2 - 4x + 9$ 
2 polynomial1 = [9.0, -4.0, 3.0, -5.0, 1.0]
3
4 #  $2x^3 + 4x^2 - 9$ 
5 polynomial2 = [-9.0, 0.0, 4.0, 2.0]
6 polynomial3 = add_polynomials(polynomial1, polynomial2)
7 print('Sum =', polynomial3)
```

See the problem?

Exercise 3 Dealing with polynomials of different degrees

Working Space

Can you fix the function `add_polynomials` to handle polynomials of different degrees?

Here is a hint: In Python, there is a `max` function that returns the largest of the numbers it is passed.

```
1 biggest = max(5,7)
```

Here `biggest` would be set to 7.

Here is another hint: If you have an array `mylist`, `i`, a non-negative integer, is only a legit index if `i < len(mylist)`.

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1.3 Scalar multiplication of polynomials

If you multiply a polynomial with a number, the distributive property applies:

$$(3.1)(2x^2 + 3x + 1) = (6.2)x^2 + (9.3)x + 3.1$$

(When we are talking about things that are more complicated than a number, we use the word *scalar* to mean “Just a number”. This is the product of a scalar and a polynomial.)

In `add_polynomials.py`, add a function to that multiplies a scalar and a polynomial:

```
1 def scalar_polynomial_multiply(s, pn):
2     result = []
3     for coefficient in pn:
4         result.append(s * coefficient)
5     return result
```

Somewhere near the end of the program, test this function:

```
1 polynomial4 = scalar_polynomial_multiply(5.0, polynomial1)
2 print('Scalar product =', polynomial_to_string(polynomial4))
```

Exercise 4 Subtract polynomials in Python

Now, implement a function that does subtraction using `scalar_polynomial_multiply` and `add_polynomials`.

Working Space

It should look like this:

```
1 def subtract_polynomial(a, b):
2     ...Your code here...
3
4 polynomial5 = [9.0, -4.0, 3.0,
5     ↪ -5.0]
6 polynomial6 = [-9.0, 0.0, 4.0, 2.0,
7     ↪ 1.0]
8 polynomial7 =
9     ↪ subtract_polynomial(polynomial5,
10    ↪ polynomial6)
11 print('Difference =',
12    ↪ polynomial_to_string(polynomial7))
```

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This is a draft chapter from the Kontinua Project. Please see our website (<https://kontinua.org/>) for more details.

Answers to Exercises

Answer to Exercise 1 (on page 2)

$$3x^3 - 7x^2 + x - 18 \text{ and } 3x^5 - 7x^3 + x^2 - 12$$

Answer to Exercise 2 (on page 3)

$$x^3 - 3x^2 + 5x \text{ and } x^5 - 3x^3 + 5x^2 - 2x + 6$$

Answer to Exercise 3 (on page 4)

```
1 def add_polynomials(a, b):
2     degree_of_result = max(len(a), len(b))
3     result = []
4     for i in range(degree_of_result):
5         if i < len(a):
6             coefficient_a = a[i]
7         else:
8             coefficient_a = 0.0
9
10        if i < len(b):
11            coefficient_b = b[i]
12        else:
13            coefficient_b = 0.0
14
15        result.append(coefficient_a + coefficient_b)
16    return result
```

Answer to Exercise 4 (on page 6)

```
1 def subtract_polynomial(a, b):
2     neg_b = scalar_polynomial_multiply(-1.0, b)
3     return add_polynomials(a, neg_b)
```




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